

reminders: ① vector addition & subtraction

② $\vec{a}$ $\xrightarrow{\text{decomposition}}$ $\begin{cases} \vec{a}_t \rightarrow \frac{dv}{dt} \\ \vec{a}_r \rightarrow \frac{d^2r}{dt^2} \end{cases}$

	linear constant v	linear accelerating	uniform circular	general
$\vec{a}_t$	0	$\neq 0$	0	$\neq 0 \left(\frac{dv_A}{dt} \right)$
$\vec{a}_r$	0	0	$\neq 0 \left(\frac{v^2}{r} \right)$ ↑ named " a_c "	$\neq 0 \left(\frac{v_A^2}{r_A} \right)$

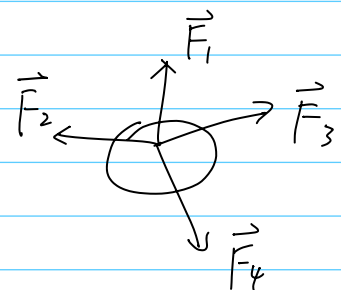
there is only one acceleration in reality.
 math operation: helps us understand the physical meaning, & to make problem easier.
 one of many ways to decompose

review

net force exerted on that object $\leftarrow$ $\uparrow$ mass of that object

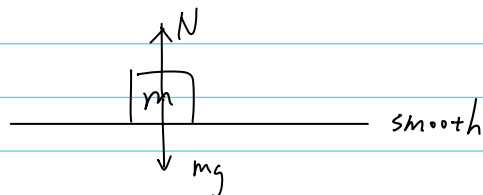
$$\vec{F}_{\text{net}} = m\vec{a}$$

$$\vec{F}_{\text{net}} = \sum_i \vec{F}_i = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \dots = m\vec{a}$$



- If at rest or constant $\vec{v}$, $\vec{a} = 0$, "equilibrium"
- decompose into two perpendicular directions and analyze $F=ma$ on each direction.

e.g. $\rightarrow \vec{v} = \text{constant}$



analyze the forces.

$\rightarrow$ decompose into horizontal & vertical directions, then analyze each.

In x , $a_x = 0$, $\sum \vec{F}_x = m a_x = 0$ $\frac{d\vec{v}}{dt} = 0$
 $\leftarrow$ 0 no force

In y , $a_y = 0$, $\sum \vec{F}_y = N - mg = m a_y = 0 \Rightarrow N = mg$

N is determined by mg . $m \uparrow N \uparrow$.

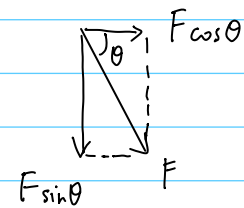
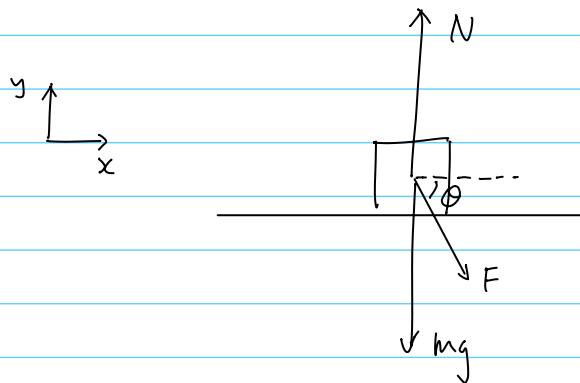
N can be as large as long as they don't break.

Note : 1. the vector form, e.g. $\vec{F}$, contains both magnitude and direction.
e.g. $\vec{N} + m\vec{g} = m\vec{a}$ the "+" sign is for vector addition.

2. define your coordinates and positive directions. then you can use the symbol without $\rightarrow$ to note the components, which are scalars. They can be + or -, meaning they point toward or opposite to the positive direction you defined.

e.g. I write $N - mg$ by assuming upward as the positive direction, and $N > 0$, $mg > 0$.

Now, press block with force F pointing angle θ below horizon,
what is $\vec{N}$? what is $\vec{a}$? start at rest, $\vec{r}(t)$?



★ Normal force is always, normal (perpendicular) to the interface.

$$\Sigma F_y = N - F \sin \theta - mg = m a_y \quad a_y = 0$$

$$N = mg + F \sin \theta$$

$$\therefore \vec{N} = (mg + F \sin \theta) \hat{j}$$

In this case, only y component.

$$\Sigma F_x = F \cos \theta \quad \text{smooth, no friction.} = m a_x$$

$$a_x = \frac{F \cos \theta}{m}$$

$$\vec{a} = a_x \hat{i} + a_y \hat{j} = \frac{F \cos \theta}{m} \hat{i}$$

constant acceleration motion.

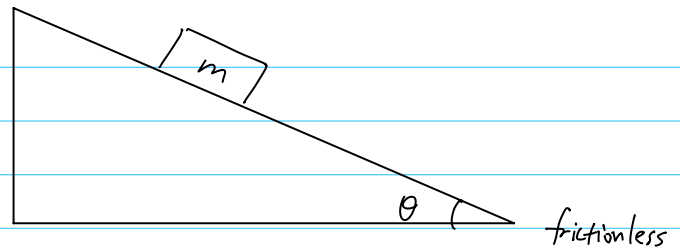
$$\vec{r} = \frac{1}{2} \vec{a} t^2 + \vec{v}_i t + \vec{r}_i = \frac{F \cos \theta t^2}{2m} \hat{i}$$

* Another example: projectile motion $\begin{cases} x: \text{no } F_{x \text{ net}}, \text{ therefore } a_x = 0 \\ y: F_{y \text{ net}} = -mg, \text{ therefore } a_y = -g \text{ (set upward positive)} \end{cases}$

A few representative models:

1) inclined plane

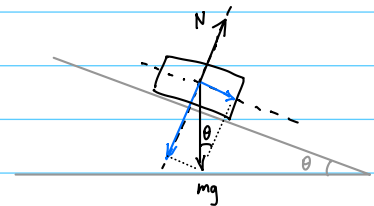
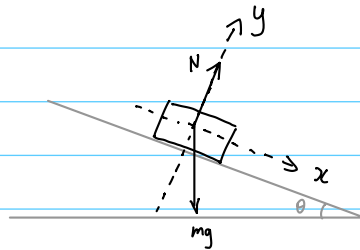
Given m, θ, g , find N, a



★ You can choose any rectangular coordinate system, but it's easier to put x along surface, so that I don't need to decompose $\vec{a}$.

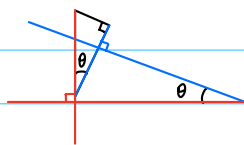
(1) set up coordinate
draw free-body diagram

(2) decompose along x and y .

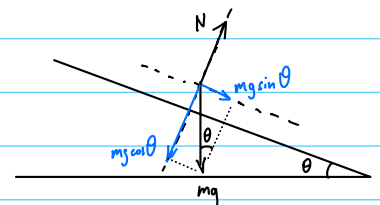


(3) Find the straight triangle
find angle θ . blue $\perp$ blue
red $\perp$ red

(4) Determine the components.



angles bounded by the pair
are the same.



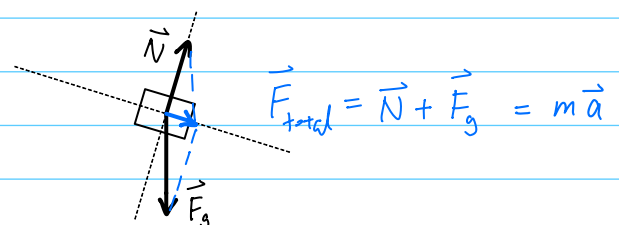
$$\begin{aligned}\vec{N} &= N \hat{j} \\ \vec{F}_g &= mg \sin \theta \hat{i} - mg \cos \theta \hat{j} \\ \vec{a} &= a \hat{i}\end{aligned}$$

$$\sum F_y = N - mg \cos \theta = 0 \Rightarrow N = mg \cos \theta$$

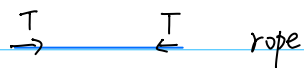
$$\sum F_x = mg \sin \theta = ma \Rightarrow a = g \sin \theta \quad \text{independent of } m!$$

$$\theta \rightarrow 0 \quad a \rightarrow 0$$

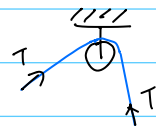
$$\theta \rightarrow 90^\circ \quad a \rightarrow g$$



2). massless frictionless pulley (fixed pulley)

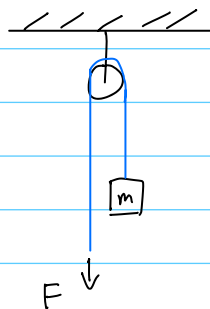


rope



rope + fixed pulley

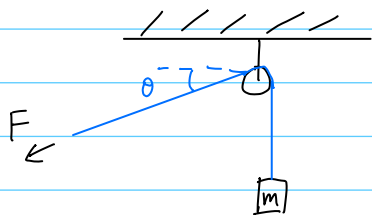
on two sides: $\vec{T}$ same magnitude, pointing towards string
 $\vec{x}, \vec{v}, \vec{a}$ same magnitude, along same direction of string.



System in equilibrium, $F = ?$



$$F = mg$$



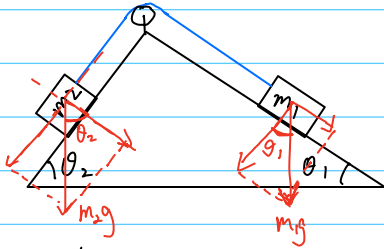
System in equilibrium, $F = ?$



$$F = mg$$

doesn't depend on θ .

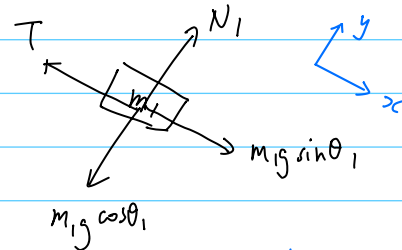
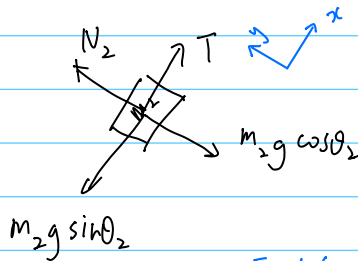
3) incline plane + pulley



frictionless

Given $m_1, m_2, \theta_1, \theta_2, g$,

find a (how will they move?)



I define the $+x$ directions like this, so that a takes the same sign.

In the rope direction (x) :

I don't know which direction a is yet.

I let $\vec{a} = a \hat{i}$, in the end, if I find

$a > 0$ (or $a < 0$) that means $\vec{a}$ is along $+x$ (or $-x$) direction.

$$\text{For 2: } T - m_2 g \sin \theta_2 = m_2 a \quad (1)$$

$$\text{For 1: } m_1 g \sin \theta_1 - T = m_1 a \quad (2)$$

a is same

$$(1) : T = m_2 a + m_2 g \sin \theta_2 \quad \rightarrow \text{put in (2)}$$

$$m_1 g \sin \theta_1 - m_2 a - m_2 g \sin \theta_2 = m_1 a$$

$$m_1 g \sin \theta_1 - m_2 g \sin \theta_2 = (m_1 + m_2) a$$

$$a = g \frac{m_1 \sin \theta_1 - m_2 \sin \theta_2}{m_1 + m_2}$$

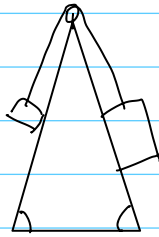
If $m_1 \sin \theta_1 > m_2 \sin \theta_2 \Rightarrow a > 0 \Rightarrow m_1$ slides down

If $m_1 \sin \theta_1 < m_2 \sin \theta_2 \Rightarrow a < 0 \Rightarrow m_2$ slides down

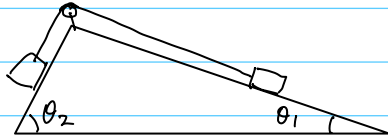
If $m_1 \sin \theta_1 = m_2 \sin \theta_2 \Rightarrow a = 0 \Rightarrow$ system in equilibrium

(doesn't depend on rope length on each side)

Imagine

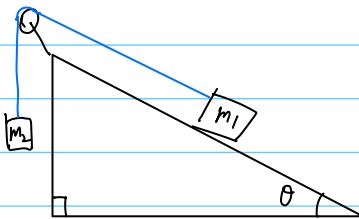


If $\theta_1 = \theta_2$,
the heavier block slides down



If $m_1 = m_2$,
 $\theta_2 > \theta_1$,
 $\sin \theta_2 > \sin \theta_1$,
 m_1 slides down

Special case $\theta_2 = 90^\circ$



$$\begin{cases} T - m_2 g = m_2 a \\ m_1 g \sin \theta_1 - T = m_1 a \end{cases}$$

$$m_1 g \sin \theta - m_2 a + m_2 g = m_1 a$$

$$a = g \cdot \frac{m_1 \sin \theta + m_2}{m_1 + m_2}$$

5.8

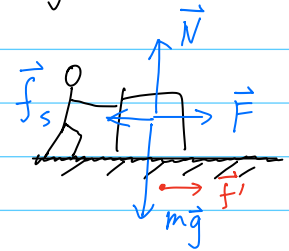
Friction

- Static friction: when the object is stationary $\vec{v} = 0$

e.g. I push a box with $\vec{F}$, but it's not moving

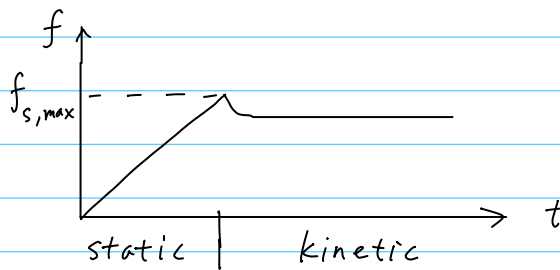
direction: opposite to the motion tendency.

magnitude: since $\frac{d\vec{v}}{dt} = 0$, $|\vec{f}_s| = |\vec{F}|$



* Box is also pushing the ground with $\vec{f}'$, $\vec{f}'$ and $\vec{F}$ are a
action - reaction pair.

If I start from $\vec{F} = 0$ and increase $\vec{F}$, while the box is not moving, $f_s = F$, eventually when $f_s = f_{s, \max}$, the box starts to move.
(maximum static friction).

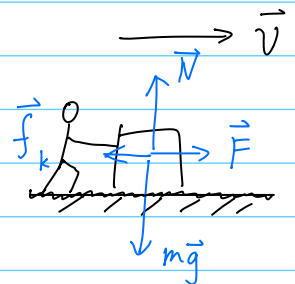


- kinetic friction: when the object is moving

$$f_k = \mu_k N$$

↑
coefficient of kinetic friction

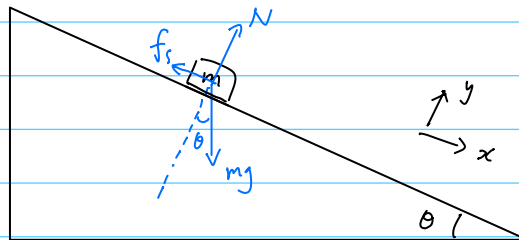
- $f_k \propto N$
- f_k is independent of v



$$(f_{s, \max} = \mu_{s, \max} N, \quad \mu_{s, \max} > \mu_k)$$

model: inclined plane with friction

1) Given m , $\mu_{s,\max}$, largest θ without block slipping?



in x : $mg \sin \theta - f_s = ma_x = 0$.

$f_s = mg \sin \theta$, when θ increases f_s increases,

$f_s \leq f_{s,\max}$. beyond this limit, block will slip

$$mg \sin \theta \leq \mu_{s,\max} \cdot N$$

in y : $N - mg \cos \theta = ma_y = 0$

$$\cancel{mg} \sin \theta \leq \mu_{s,\max} \cdot \cancel{mg} \cos \theta$$

$$\tan \theta \leq \mu_{s,\max}$$

critical angle is $\tan^{-1} \mu_{s,\max}$

2) Given θ , μ_k , if block is sliding, what is a ?

$$\cancel{mg} \sin \theta - \mu_k \cancel{mg} \cos \theta = ma$$

$$a = g \cdot (\sin \theta - \mu_k \cos \theta)$$